Structural Foundations of the Critical Coherence Index

Definition, Physical Proxy, and Empirical Scaling

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Abstract

The Critical Coherence Index (CCI) has become a central observable in the structural-selection programme for nonlinear systems. Across the existing paper series, however, three distinct layers have often appeared in close succession: the formal definition of the CCI, its physical proxy representation in nonlinear field systems, and its empirical entropy–information scaling law. The aim of the present paper is to separate these layers cleanly and to place the CCI on a canonical theoretical footing.

We define the CCI first as a dimensionless diagnostic ratio between destabilising and coherence-preserving structural contributions. We then show how, in nonlinear field systems, this abstract definition admits a measurable proxy in terms of coarse-grained histogram activity and nearest-neighbour mutual information. Finally, we distinguish this proxy from the later empirical scaling relation involving an effective exponent α, which should be understood as a phenomenological fit rather than as part of the definition itself.

This clarification yields a three-level interpretation: formal definition, physical representation, and empirical scaling. Within this hierarchy, the CCI functions as a reduced instability observable, while the structural free energy remains the higher-level selection functional. The result is not a new numerical claim, but a conceptual consolidation of the series. It is intended to make the CCI mathematically sharper, physically more transparent, and easier to transport across field systems, optimisation settings, and future compactification benchmarks.

1 Introduction

The Critical Coherence Index (CCI) was introduced in the structural-selection programme as a dimensionless instability diagnostic for nonlinear systems. In the early papers, its role was intuitive and operational: low values indicated ordered configurations, intermediate values signalled critical behaviour, and high values were associated with chaotic or instability-dominated dynamics. Later papers connected the CCI to coarse-grained activity, structural information, scaling behaviour across dimension, and regime classification.

This development produced a fruitful but increasingly layered picture. Three different statements began to travel under the same symbol:

1. a *formal structural definition*,
2. a *physical proxy representation* in nonlinear field systems,
3. an *empirical scaling law* extracted from ensembles.

Those three statements are mutually compatible, but they are not identical. Conflating them weakens the framework unnecessarily. In particular, the formal definition should remain valid even when a given proxy fails, and an empirical scaling law should never be mistaken for the definition of the observable itself.

The purpose of this paper is therefore narrow but important: to provide a canonical theoretical layering of the CCI. The result is a clean three-step hierarchy:

$$\text{definition} \longrightarrow \text{physical proxy} \longrightarrow \text{empirical scaling.}$$

This clarification matters for the entire series. It stabilises the interpretation of Papers 04–17, sharpens the relation between the CCI and the structural free energy, and clarifies how later benchmark papers such as the minimal ϕ^4 test case should be read.

2 Why a Canonical Definition is Needed

The existing series already contains the ingredients of a mature CCI theory, but they have not yet been consolidated into one canonical block. This produces three recurring ambiguities.

First ambiguity: definition versus interpretation. The CCI was originally introduced as a structural instability ratio, but in later work it was often discussed through the more concrete language of entropy proxies and mutual information. That language is physically useful, but it belongs to interpretation, not to the abstract definition itself.

Second ambiguity: proxy versus scaling. The approximation

$$\mathcal{C}_{\mathrm{CCI}} \approx \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}} + \varepsilon}$$

already has a different status from the later scaling form

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}.$$

The first is a physically motivated representation in a particular class of systems; the second is an empirical fit across ensembles. Treating them as the same statement obscures what has actually been established.

Third ambiguity: local observable versus global selection functional. The CCI is a reduced instability observable. The structural free energy $\mathcal{F}_{\mathrm{struct}}$ is a higher-level selection functional. They are strongly related, but not identical. A canonical definition of the CCI helps preserve that distinction.

3 Canonical Definition of the Critical Coherence Index

The CCI is introduced as a dimensionless diagnostic of structural stability in nonlinear systems. Its definition is organised in three layers: a formal definition, a physical proxy, and an empirical scaling relation.

3.1 Formal Definition

We define the CCI as a ratio between destabilising and stabilising structural contributions:

$$\boxed{\mathcal{C}_{\mathrm{CCI}} := \frac{\Gamma_{\mathrm{inst}} \cdot \Gamma_{\mathrm{prod}} \cdot (1 + \kappa\, U_{\mathrm{struct}})}{\Gamma_{\mathrm{coh}} + \Gamma_{\mathrm{cons}} + \Gamma_{\mathrm{corr}} + \Gamma_{\mathrm{int}} + \varepsilon}} \tag{1}$$

Here:

- Γ_{inst} denotes dynamical instability,
- Γ_{prod} measures structural or entropic activity,

- U_{struct} quantifies structural imbalance,
- Γ_{coh} measures coherence,
- Γ_{cons} enforces conservation consistency,
- Γ_{corr} captures correlation structure,
- Γ_{int} measures internal integrity through a defect-sensitive score,
- $\kappa \geq 0$ weights the imbalance contribution,
- $\varepsilon > 0$ is a denominator regularisation scale.

The quantities entering (1) are assumed to be dimensionless and non-negative after domain-appropriate normalisation. In this sense, the CCI is a formal diagnostic architecture rather than a single model-specific formula. What changes from domain to domain is the explicit realisation of the Γ sectors, not the structural form of the definition.

The role of ε should be understood narrowly. It is not an additional physical sector, but a small denominator floor that prevents spurious divergences when the stabilising channels become very small. In practical implementations, a natural choice is

$$\varepsilon = \eta_\varepsilon \, D_{\text{ref}}, \qquad 0 < \eta_\varepsilon \ll 1,$$

where D_{ref} is a representative stabilising scale, for example a typical or smallest non-zero value of $\Gamma_{\text{coh}} + \Gamma_{\text{cons}} + \Gamma_{\text{corr}} + \Gamma_{\text{int}}$ over the ensemble under study.

3.2 Why This Algebraic Form?

The choice of a multiplicative numerator and an additive denominator is not mathematically unique, but it is not arbitrary either. It is a minimal form chosen to satisfy three structural requirements.

1. **Joint activation in the numerator.** The factor

   $$\Gamma_{\text{inst}}\Gamma_{\text{prod}}$$

   ensures that the CCI remains small if either instability or activity is absent. A configuration that is highly active but not destabilising, or destabilising in a purely latent sense but dynamically inactive, should not automatically count as structurally critical.

2. **Modulation by imbalance.** The factor

   $$1 + \kappa U_{\text{struct}}$$

   does not create instability on its own. It modulates the basic instability–activity product by structural imbalance, so that imbalance amplifies an already active instability rather than replacing it.

3. **Accumulation of stabilising channels in the denominator.** The additive denominator

   $$\Gamma_{\text{coh}} + \Gamma_{\text{cons}} + \Gamma_{\text{corr}} + \Gamma_{\text{int}} + \varepsilon$$

   encodes the idea that coherence-preserving mechanisms accumulate in parallel. Increasing any one of these channels should increase the stabilising capacity of the configuration even if the others remain fixed.

Alternative aggregations are possible and may become relevant in future refinements. The present choice should therefore be read as a minimal monotone ansatz rather than as a uniquely derived theorem.

3.3 Domain-Independent Interpretation

Equation (1) says that the CCI grows when destabilising activity outpaces the mechanisms that preserve coherence, conservation, correlation, and internal integration. It is therefore a structural analogue of Reynolds-type instability numbers: not a measure of one ingredient in isolation, but of the balance between competing tendencies.

This balance yields the qualitative regime interpretation used throughout the series:

- low CCI: ordered or strongly coherent configurations,
- intermediate CCI: critical or transition-like configurations,
- high CCI: instability-dominated or chaotic configurations.

These regime labels belong to the interpretation of the index, not to its formal definition. Their numerical thresholds may depend on the model, the ensemble, and the normalisation used for the component sectors.

3.4 Illustrative Toy Example

To make the formal definition more concrete, consider three purely illustrative normalised states with $\kappa = 1$ and $\varepsilon = 10^{-6}$:

State	Γ_{inst}	Γ_{prod}	U_{struct}	denominator	$\mathcal{C}_{\text{CCI}}$	regime
Ordered	0.2	0.3	0.1	4.0	0.0165	ordered
Critical	0.8	0.9	0.2	2.8	0.3086	critical
Chaotic	0.95	0.90	0.4	0.5	2.3940	chaotic

Here the denominator abbreviates

$$\Gamma_{\text{coh}} + \Gamma_{\text{cons}} + \Gamma_{\text{corr}} + \Gamma_{\text{int}} + \varepsilon.$$

The table is not a new physical result; it is only a diagnostic illustration. Its role is to show that (1) behaves as intended: low joint instability and activity lead to small CCI, intermediate balances lead to moderate CCI, and strong active instability combined with weak stabilising channels produces large CCI.

3.5 Canonical Nonlinear-Field Closure of the Γ Sectors

For nonlinear field systems, the abstract sectors in (1) can be closed canonically by explicit dimensionless residual functionals. A minimal field-theoretic realisation is:

$$\Gamma_{\mathrm{coh}}[\Phi] := 1 - \frac{\sum_i \left\langle \left(\frac{\delta S}{\delta \Phi_i}\right)^2 \right\rangle}{\Lambda_R + \sum_i \left\langle \left(\frac{\delta S}{\delta \Phi_i}\right)^2 \right\rangle}, \tag{2}$$

$$\Gamma_{\mathrm{cons}}[\Phi] := 1 - \frac{\langle (\nabla_\mu T^{\mu\nu})(\nabla_\rho T^\rho{}_\nu)\rangle}{\Lambda_T + \langle (\nabla_\mu T^{\mu\nu})(\nabla_\rho T^\rho{}_\nu)\rangle}, \tag{3}$$

$$\Gamma_{\mathrm{corr}}[\Phi] := \frac{I_{\mathrm{nn}}}{I_0 + I_{\mathrm{nn}}}, \tag{4}$$

$$\Gamma_{\mathrm{prod}}[\Phi] := \frac{\dot{S}^{(+)}_{\mathrm{cg}}}{S_0 + \dot{S}^{(+)}_{\mathrm{cg}}}, \tag{5}$$

$$\Gamma_{\mathrm{int}}[\Phi] := 1 - \frac{\mathcal{I}_{\mathrm{int}}[\Phi]}{\Lambda_{\mathrm{int}} + \mathcal{I}_{\mathrm{int}}[\Phi]}, \tag{6}$$

$$\Gamma_{\mathrm{inst}}[\Phi] := \frac{\mathcal{I}_{\mathrm{inst}}[\Phi]}{\Lambda_{\mathrm{inst}} + \mathcal{I}_{\mathrm{inst}}[\Phi]}. \tag{7}$$

Here Λ_R, Λ_T, Λ_{int}, and Λ_{inst} are positive normalisation scales, while I_0 and S_0 set the information and activity reference scales. The brackets $\langle\cdot\rangle$ denote the domain-appropriate averaging procedure, for example a spatial, ensemble, or coarse-grained lattice average.

The corresponding structural-imbalance factor is then defined canonically by the dispersion of the five normalised structural sectors Γ_{coh}, Γ_{cons}, Γ_{prod}, Γ_{corr}, and Γ_{int}:

$$\mu_\Gamma := \frac{1}{5} \sum_{a\in\{\mathrm{coh,cons,prod,corr,int}\}} \Gamma_a, \qquad U_{\mathrm{struct}} := \left[\frac{1}{5} \sum_{a\in\{\mathrm{coh,cons,prod,corr,int}\}} (\Gamma_a - \mu_\Gamma)^2\right]^{1/2}. \tag{8}$$

The exclusion of Γ_{inst} from (8) is intentional. Instability acts as a direct driver in the numerator of (1), whereas U_{struct} measures mismatch among the remaining normalised structural sectors. If Γ_{inst} were included in the dispersion measure, the same instability channel would act both as a direct driver and as an indirect imbalance amplifier, effectively double-counting its influence. For this reason, the five-sector definition in (8) is taken as canonical here. A six-sector variant, including Γ_{inst} in the dispersion, may still be useful as a robustness or sensitivity test in future work.

Two of the sector inputs require one further level of operational specification.

Instability sector. The quantity $\mathcal{I}_{\mathrm{inst}}[\Phi]$ should be any non-negative diagnostic of active instability. In perturbative continuum settings, it may be taken as a local perturbation-growth rate or Lyapunov-type indicator. In the minimal Hamiltonian lattice implementation used in the early ϕ^4 studies, a practical default is

$$\mathcal{I}_{\mathrm{inst}}[\Phi] = \langle \pi^2 \rangle,$$

which yields the canonical saturation form

$$\Gamma_{\mathrm{inst}}[\Phi] = \frac{\langle \pi^2 \rangle}{\Lambda_\pi + \langle \pi^2 \rangle}.$$

Integrity sector. The quantity $\mathcal{I}_{\text{int}}[\Phi]$ measures internal loss of configuration integrity. In the scalar-lattice benchmark, a simple choice is

$$\mathcal{I}_{\text{int}}[\Phi] = \left\langle \max\big(0, |\phi| - \phi_{\max}\big)^2 \right\rangle ,$$

which penalises excursions beyond the admissible field-amplitude window. In more general field systems, the same slot can be occupied by a normalised coupling-integrity or boundedness diagnostic. The raw quantity $\mathcal{I}_{\text{int}}$ is therefore a defect variable, while Γ_{int} is the corresponding integrity-preserving support score obtained after normalised inversion.

With (2)–(7) and (8), the formal definition (1) becomes a closed nonlinear-field expression rather than a merely schematic template. The remaining model freedom lies not in whether the sectors exist, but in the choice of the most faithful concrete diagnostics inside $\mathcal{I}_{\text{inst}}$ and $\mathcal{I}_{\text{int}}$ for a given field content.

3.6 Axiomatic Cross-Domain Closure

The same closure logic can be extended beyond nonlinear field systems if one distinguishes between *sector roles*, which are universal, and *primitive diagnostics*, which are domain-dependent. Let D be a target domain with state space $\mathcal{X}_D$, and let $a \in \{\text{inst}, \text{prod}, \text{coh}, \text{cons}, \text{corr}, \text{int}\}$ label the six CCI sectors. For each sector, choose:

- a non-negative primitive diagnostic $r_a^{(D)}[x]$ on $\mathcal{X}_D$,
- a positive normalisation scale $\Lambda_a^{(D)}$,
- an orientation flag $\sigma_a^{(D)} \in \{+1, -1\}$.

The orientation flag indicates whether larger raw values mean stronger sector support ($\sigma = +1$) or larger sector defect ($\sigma = -1$). The universal closure operator is then

$$\mathcal{N}_\sigma(r; \Lambda) := \begin{cases} \dfrac{r}{\Lambda + r}, & \sigma = +1, \\ \dfrac{\Lambda}{\Lambda + r}, & \sigma = -1, \end{cases} \qquad r \geq 0,\ \Lambda > 0, \tag{9}$$

and the domain-specific sector score is defined by

$$\Gamma_a^{(D)}[x] := \mathcal{N}_{\sigma_a^{(D)}}\left(r_a^{(D)}[x]; \Lambda_a^{(D)}\right). \tag{10}$$

This gives a domain-universal closure rule: the Γ sectors are closed not by identical formulas in every domain, but by identical *normalisation and orientation principles*. The cross-domain admissibility conditions are:

1. **Dimensionless boundedness.** Each $\Gamma_a^{(D)}$ must lie in $[0, 1]$ after normalisation.
2. **Symmetry respect.** Each primitive $r_a^{(D)}$ must be invariant, or at least covariant in a controlled way, under the natural symmetries of the target domain.
3. **Orientation fidelity.** The chosen sign $\sigma_a^{(D)}$ must guarantee that larger $\Gamma_a^{(D)}$ always means more sector support in the intended CCI role.
4. **Non-redundancy.** The six sectors should not collapse into trivial duplicates of one another over generic ensembles.
5. **Computability.** The primitives $r_a^{(D)}$ must be evaluable from the actual data or state description used in the model.

In this sense, universality is achieved at the level of the closure schema rather than at the level of one rigid microscopic formula.

Optimisation example. For a constrained optimisation state (x, λ) with objective f, constraints $c(x) = 0$, and Lagrangian $\mathcal{L}(x, \lambda)$, one possible instantiation is:

$$\begin{aligned}
r_{\text{coh}}^{(\text{opt})} &= \|\nabla_x \mathcal{L}(x, \lambda)\|^2, \qquad \sigma_{\text{coh}}^{(\text{opt})} = -1, \\
r_{\text{cons}}^{(\text{opt})} &= \|c(x)\|^2, \qquad \sigma_{\text{cons}}^{(\text{opt})} = -1, \\
r_{\text{inst}}^{(\text{opt})} &= \max\big(0, -\lambda_{\min}(\nabla_x^2 f)\big), \qquad \sigma_{\text{inst}}^{(\text{opt})} = +1, \\
r_{\text{prod}}^{(\text{opt})} &= \|\Delta x\|^2 / \Delta t^2, \qquad \sigma_{\text{prod}}^{(\text{opt})} = +1, \\
r_{\text{corr}}^{(\text{opt})} &= \mathcal{C}_G(x), \qquad \sigma_{\text{corr}}^{(\text{opt})} = +1, \\
r_{\text{int}}^{(\text{opt})} &= \mathcal{B}_{\text{overflow}}(x), \qquad \sigma_{\text{int}}^{(\text{opt})} = -1.
\end{aligned} \tag{11}$$

Here $\mathcal{C}_G$ may be an interaction-graph coherence score, while $\mathcal{B}_{\text{overflow}}$ measures barrier or boundedness violation.

Compactification example. For an effective compactification background $\mathcal{B}$, one may analogously choose

$$\begin{aligned}
r_{\text{coh}}^{(\text{comp})} &= \left\| \frac{\delta S_{\text{eff}}}{\delta \varphi} \right\|^2 \text{ or beta-function residuals}, \qquad \sigma_{\text{coh}}^{(\text{comp})} = -1, \\
r_{\text{cons}}^{(\text{comp})} &= |\Delta Q_{D3}|^2 + \mathcal{R}_{\text{Bianchi}}^2 + \mathcal{R}_{\text{anom}}^2, \qquad \sigma_{\text{cons}}^{(\text{comp})} = -1, \\
r_{\text{inst}}^{(\text{comp})} &= \max(0, -m_{\min}^2) + \gamma_{\text{decay}}, \qquad \sigma_{\text{inst}}^{(\text{comp})} = +1, \\
r_{\text{prod}}^{(\text{comp})} &= \mathcal{A}_{\text{uplift/backreaction}}, \qquad \sigma_{\text{prod}}^{(\text{comp})} = +1, \\
r_{\text{corr}}^{(\text{comp})} &= \mathcal{C}_{\text{moduli-flux-brane}}, \qquad \sigma_{\text{corr}}^{(\text{comp})} = +1, \\
r_{\text{int}}^{(\text{comp})} &= \mathcal{D}_{\text{EFT control}}, \qquad \sigma_{\text{int}}^{(\text{comp})} = -1.
\end{aligned} \tag{12}$$

This is not yet a unique compactification formula, but it is no longer merely a verbal wish list. It is an operational translation dictionary from the universal CCI sectors to explicit compactification diagnostics.

The optimisation and compactification examples above illustrate how the closure schema can be transported across domains. They are not yet validated benchmark models. Concrete numerical or phenomenological tests of these instantiations belong to companion domain-specific papers.

4 Physical Proxy Representation in Nonlinear Field Systems

In nonlinear field systems, the CCI admits a physically motivated approximation in terms of coarse-grained histogram activity and structural information:

$$\mathcal{C}_{\text{CCI}} \approx \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}} + \varepsilon} \tag{13}$$

where:

- $\dot{S}_{\text{cg}}^{(+)}$ is a coarse-grained histogram-activity proxy derived from the time variation of the field-value distribution,
- I_{nn} is the nearest-neighbour mutual information.

This proxy was developed to give the formal CCI a measurable representation in nonlinear field dynamics. It captures a simple physical idea: instability grows when coarse-grained reorganisation outpaces the field's capacity to retain structural correlations.

Two cautions are essential.

First, (13) is not the definition of the CCI. It is a physically motivated representation that holds in a specific class of systems.

Second, $\dot{S}_{\mathrm{cg}}^{(+)}$ should not automatically be interpreted as a literal thermodynamic entropy production rate. In closed Hamiltonian systems such as the ϕ^4 benchmark used in the companion studies, the true phase-space entropy is conserved. The quantity $\dot{S}_{\mathrm{cg}}^{(+)}$ is better understood as a coarse-grained activity or histogram-turnover proxy. The CCI can still be physically grounded through this proxy, but only with this caveat stated explicitly.

5 Empirical Scaling Relation

Across ensembles of configurations, the CCI follows an empirical scaling relation of the form

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}} \tag{14}$$

where α is an effective scaling exponent that depends on system dimension and structural complexity.

This relation belongs to a third layer of the theory. It is neither the formal definition nor the physical proxy itself. Rather, it is a phenomenological ensemble law extracted from numerical studies.

The series already suggests the following qualitative pattern:

- within the present ϕ^4-based series, low-dimensional systems admit a relatively stable effective exponent,
- higher-dimensional systems exhibit broad exponent plateaus,
- the resulting degeneracy signals a breakdown of one-parameter scaling rather than the breakdown of the CCI itself.

This should not yet be read as a universal claim about all nonlinear systems. The current evidence is strongest for the ϕ^4 benchmark family treated in the companion scaling papers. Detailed visual evidence for the exponent plateaus belongs to those dedicated papers; the present note only fixes the theoretical status of the scaling layer relative to the formal definition and the physical proxy.

This is important. The CCI can remain meaningful even when simple power-law collapse fails. In that sense, plateau degeneracy is a property of the scaling representation, not of the formal observable.

6 Interpretation: Definition, Proxy, and Scaling

The three levels should be clearly distinguished:

- Eq. (1) is the **definition**,
- Eq. (13) is a **physical approximation**,
- Eq. (14) is an **empirical scaling law**.

This separation ensures that the CCI remains well-defined independently of any particular modelling assumptions, while still admitting physically interpretable and measurable realisations.

The hierarchy may therefore be summarised as

$$\boxed{\begin{array}{c}\text{formal instability observable}\\ \downarrow\\ \text{measurable field-theory proxy}\\ \downarrow\\ \text{ensemble scaling representation}\end{array}}$$

7 Relation to Structural Free Energy

The CCI is not identical to the structural free energy $\mathcal{F}_{\text{struct}}$, nor to the more general master functional $\mathcal{F}_{\text{master}}$ used in the broader SST framework. The latter functionals define global selection landscapes; the former is a reduced instability observable.

The cleanest conceptual picture is:

$$\text{structural diagnostics} \longrightarrow \mathcal{C}_{\text{CCI}} \longrightarrow \mathcal{F}_{\text{struct}} \subseteq \mathcal{F}_{\text{master}}.$$

In this hierarchy:

- the structural diagnostics provide the underlying state description,
- the CCI compresses that description into a reduced stability observable,
- $\mathcal{F}_{\text{struct}}$ defines the higher-level selection landscape in field systems,
- $\mathcal{F}_{\text{master}}$ extends this logic to the broadest background-selection setting.

This explains why the CCI can correlate strongly with $\mathcal{F}_{\text{struct}}$ without being identical to it. The CCI is a lower-dimensional observable; $\mathcal{F}_{\text{struct}}$ and $\mathcal{F}_{\text{master}}$ are selection functionals acting on broader configuration spaces.

In schematic form, one may write

$$\mathcal{C}_{\text{CCI}} = \mathcal{C}(\text{instability-relevant sectors}).$$

At a broader framework level, $\mathcal{F}_{\text{master}}$ may be modelled as a functional of CCI-like instability sectors together with additional balance/order sectors. This should not be read as a derived identity. It is a framework-level statement: the CCI acts as a reduced observable that approximates instability- relevant contributions to the broader structural selection functional rather than replacing it.

8 Relation to the Existing Paper Series

The present clarification reorganises the existing series rather than replacing it.

Paper 04. The CCI was introduced as a dimensionless instability ratio and used to define ordered, critical, and chaotic regimes.

Paper 05. The physical grounding of the CCI was developed through the proxy relation (13), relating the index to coarse-grained activity and structural information.

Papers 07–16. The later scaling papers investigated the ensemble-level relation (14), including dimensional trends and plateau degeneracy.

Paper 17. The predictive power paper used the CCI as an effective classifier of dynamical regime, relying on the fact that a reduced observable can still be highly informative even if it is not a complete structural functional.

Paper 22. The minimal ϕ^4 benchmark moved one level upward, working more directly with structural functionals and sectorwise selection. This should not be read as displacing the CCI. Rather, it shows that the same logic can be realised in an explicit benchmark even when the operational ranking variable is closer to $\mathcal{F}_{\text{struct}}$ than to the reduced CCI observable.

9 Regime Structure and Thresholds

The CCI naturally organises state space into ordered, critical, and chaotic regions. What remains open is whether these transitions admit universal threshold values across broad model classes.

At present, the safe statement is:

- the qualitative regime structure is robust,
- the quantitative thresholds are not yet canonically fixed,
- any proposed universal threshold hypothesis should be treated as a conjectural next step rather than as part of the present definition.

This is another advantage of the three-layer framework: the CCI can be well-defined even when threshold universality remains unresolved.

10 Falsifiability and Open Problems

The canonical formulation clarifies what future work must establish.

1. **Validation of the cross-domain closure schema.** The present paper now gives an axiomatic closure rule for optimisation, compactification, and other non-field domains. What remains open is not the schema itself, but which concrete primitive diagnostics are most faithful and least redundant in each application class. In compactification benchmarks, a natural early stress test is the consistency primitive $r_{\text{cons}}^{(\text{comp})}$, since Bianchi, tadpole, and anomaly residuals are especially sensitive to truncation choices and neglected backreaction.
2. **Proxy robustness.** The approximation (13) must be tested beyond the current nonlinear field benchmarks.
3. **Scaling scope.** The empirical law (14) may hold only in restricted ensemble regimes; its domain of validity must be mapped systematically.
4. **Relation to $\mathcal{F}_{\text{struct}}$.** The precise monotonic or functional relation between the reduced CCI and the global structural free energy remains to be derived more sharply.
5. **First-principles derivation.** The present work canonises the definition of the CCI, but it does not yet derive the observable from a deeper microscopic theory.

11 Conclusion

The main contribution of this paper is conceptual consolidation. The Critical Coherence Index is placed on a canonical three-level footing:

1. a formal definition as a dimensionless structural instability ratio,

2. a physical proxy representation in nonlinear field systems,

3. an empirical scaling law across ensembles.

This separation resolves an ambiguity that had accumulated across the paper series. The CCI is not identical to its coarse-grained entropy–information proxy, and neither of these is identical to the later empirical scaling law. Each has a distinct theoretical status.

At the same time, the abstract Γ sectors are no longer purely schematic in the nonlinear-field setting. They can now be instantiated by explicit normalised residual functionals for coherence, conservation consistency, correlation structure, activity, instability, and internal integrity. This makes the CCI not only conceptually cleaner, but also operationally more closed in the domain from which the series originally emerged.

Beyond nonlinear field systems, the paper now also provides an axiomatic cross-domain closure rule: universality is assigned to the sector roles and to their normalisation logic, while the primitive diagnostics remain domain-specific. This is the right sense in which the CCI can be made transportable across optimisation settings and compactification benchmarks without pretending that all domains share the same microscopic formula.

With this clarification, the CCI becomes a cleaner object: a defined dimensionless observable with multiple representations. That is exactly the level of precision needed if the structural-selection programme is to mature from a promising research direction into a coherent theoretical framework.

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